



Games and Immersive Environments

2025-2026
Fernando Birra
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Lecture 1

Welcome and Course Overview



Course Introduction

- A practice-oriented course that provides a broad study of the game development process, integrating fundamental game design concepts with advanced technical implementation. It covers essential areas such as game engine architecture, the 3D graphics pipeline, procedural content generation (PCG), and the core principles of Virtual and Augmented Reality.
- Utilizes a project-based learning model, the course requires students to apply theoretical formalisms—including physics-based simulation and illumination models—to the incremental development of two functional 3D games, ranging from a small individual project to a larger group effort.

Teaching Team



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The Roadmap (Lectures)

- **Introduction:** Game concepts, industry overview, and game engine architecture.
- **Game Design:** Mechanics, Aesthetics, and Technology. Storytelling and Level Design (3D modeling, sound, textures).
- **Graphics Pipeline:** 3D Graphics Hardware and the rendering pipeline.
- **Lighting & Shading:** Light-Object interaction: Local and Global illumination models.
- **Advanced Rendering:** Texture mapping applications and shadow generation.
- **Game Engines:** Scene graphs and data management. Rendering optimizations for real-time performance.
- **Physics:** Numerical methods and particle systems. Collision handling and rigid body simulation.
- **Procedural Content:** Algorithms for procedural generation (PCG).
- **Virtual Reality:** Navigation, interaction, presence, and usability.
- **Augmented Reality:** AR concepts, theory, and tracking.

The Roadmap (Labs)

- **Learn Unity Basics:** Learn the basics of Unity 3D Game Engine.
- **Individual Project:** Put your basic Unity 3D knowledge to the test by programming a complete small game with one level, mechanics, winning conditions and user interface.
- **Group Project (Stage 1):** Groups propose a game and submit their Game Design Document (includes Task assignment per team member)
- **Group Project (Stage 2):** Develop the game in an iterative process, including several checkpoints and demos with the Project Supervisor (the teacher).

Learning Outcomes

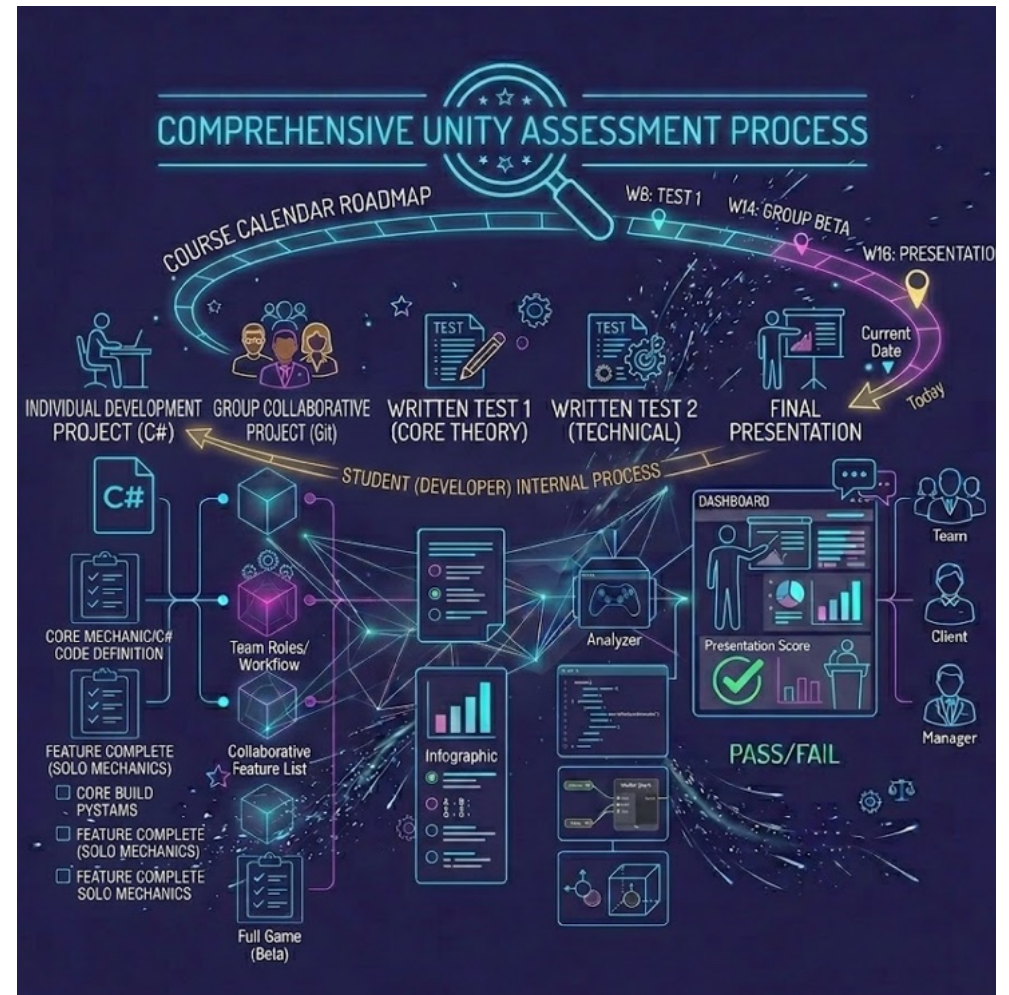
- **Game Design Foundations:** Comprehending key concepts of game design, including mechanics, aesthetics, storytelling, and the overall development process.
- **Game Engine Architecture:** Understanding the internal architecture, data management, and execution models of modern game engines and graphics hardware.
- **3D Graphics & Rendering:** Mastering the 3D graphics pipeline, including local and global illumination models and advanced real-time rendering effects.
- **Physics & Simulation:** Implementing physics-based simulations, numerical methods, and collision handling techniques essential for 3D environments.
- **Programming & Application:** Developing the ability to program specific techniques and use software components to create functional 3D games and simulations.
- **Procedural Content Generation (PCG):** Understanding and applying the main algorithms used to generate game content procedurally.
- **Immersive Technologies:** Grasping the theoretical and practical concepts of Virtual Reality (navigation/presence) and Augmented Reality.
- **Animation & GPU Optimization:** Exploring computer animation techniques and leveraging GPU resources to create high-performance visual effects.

Bibliography

Jesse Schell	<i>The Art of Game Design, A Book of Lenses</i>	2008
Jason Gregory	<i>Game Engine Architecture, 3rd Edition</i>	2018
Robert Nystrom	<i>Game Programming Patterns</i>	2014
Tomas Akenine-Möller	<i>Real-Time Rendering, 4th Edition</i>	2018
Jason Jerald	<i>The VR Book: Human-Centered Design for VR</i>	2016
Dieter Schmalstieg	<i>Augmented Reality: Principles and Practice</i>	2016



Evaluation



Component Breakdown

- [50%] A practical component (**CompL**) obtained through an Individual Project (**IP**) and a Group Project (**GP**).
 - [25%] Individual Project (**IP**) to guarantee a minimum proficiency level with the tools (Unity Engine).
 - [75%] Group Project (**GP**) (3 pax) where you can demonstrate the acquired knowledge and unleash your creativity.
- [50%] A theoretical-practical component (**CompTP**) obtained through:
 - two written tests ([50%] **T1** and [50%] **T2**) || a final Exam ([100%] **E**)

Minimum Requirements

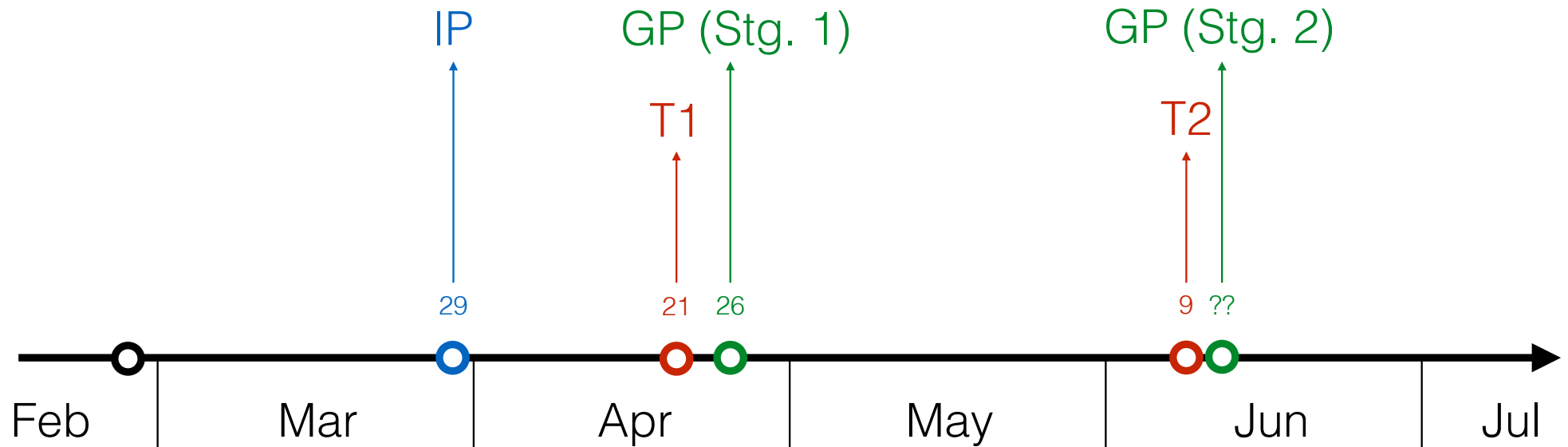
$$\mathbf{CompL} \geq 9.5 \wedge \mathbf{CompTP} \geq 0.5 \Rightarrow \mathbf{FG} \geq 9.5$$

The Whole Picture

- 2 Written tests (**T1** e **T2**) - Can be replaced by the Exam (**E**)
 - **CompTP** = $0.5 * T1 + 0.5 * T2$ | **CompTP** = **E**
 - **CompTP** ≥ 9.5 **required!**
- An Individual project (**IP**) and a group project (**GP**) in groups of **3** students
 - **CompL** = $0.25 * IP + 0.75 * GP$
 - **CompL** ≥ 9.5 **required!**
- Final Grade (**FG**)
 - **FG** = $0.5 * \text{CompTP} + 0.5 * \text{CompL}$ (if **CompTP** ≥ 9.5 && **CompL** ≥ 9.5)
 - **FG** = $\min(\text{CompTP}, \text{CompL})$ (if **CompL** < 9.5 || **CompTP** < 9.5)
- To pass, **FG** must be ≥ 9.5

Note: All grades are rounded to the first decimal place and are in the range [0,20]. Final grade is rounded to the nearest integer.

Provisional Dates



What is a Game?

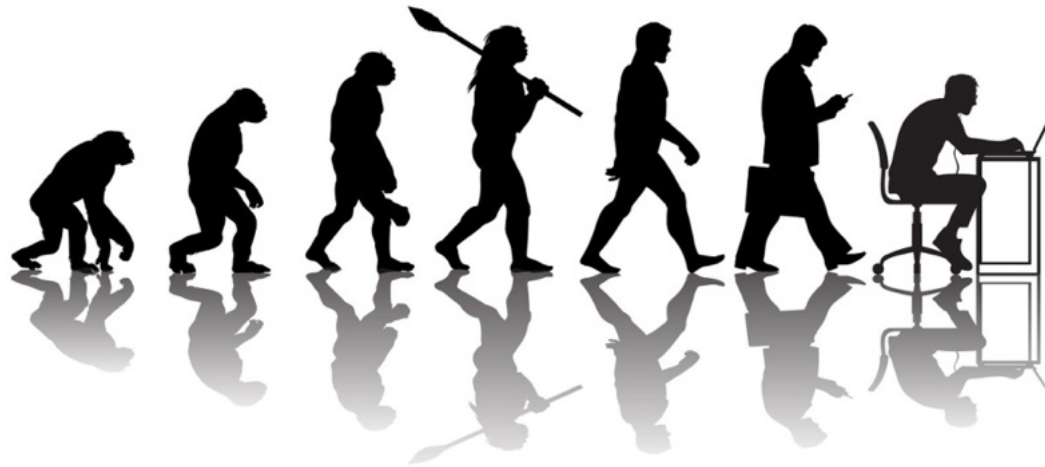
Core Concepts



What is a game? (I)

*“An **interactive** experience that provides the **player** with an increasingly **challenging sequence of patterns** which he or she **learns** and eventually **masters**.”*

in Raph Koster’s “A Theory of Fun for Game Design”



What is a game? (II)

“A game is a form of **interactive entertainment** where **players** must overcome **challenges**, by taking **actions** that are **governed by rules**, in order to meet a **victory condition**.”

in Rollings and Adams “Game Design” (2003)



Game Qualities

- A game is a problem-solving activity, approached with a playful attitude. Games...
 - ... are entered willfully
 - ... have goals
 - ... have rules
 - ... can be won and lost
 - ... are interactive
 - ... have challenge
 - ... can create their own internal value
 - ... engage players
 - ... are closed formal systems



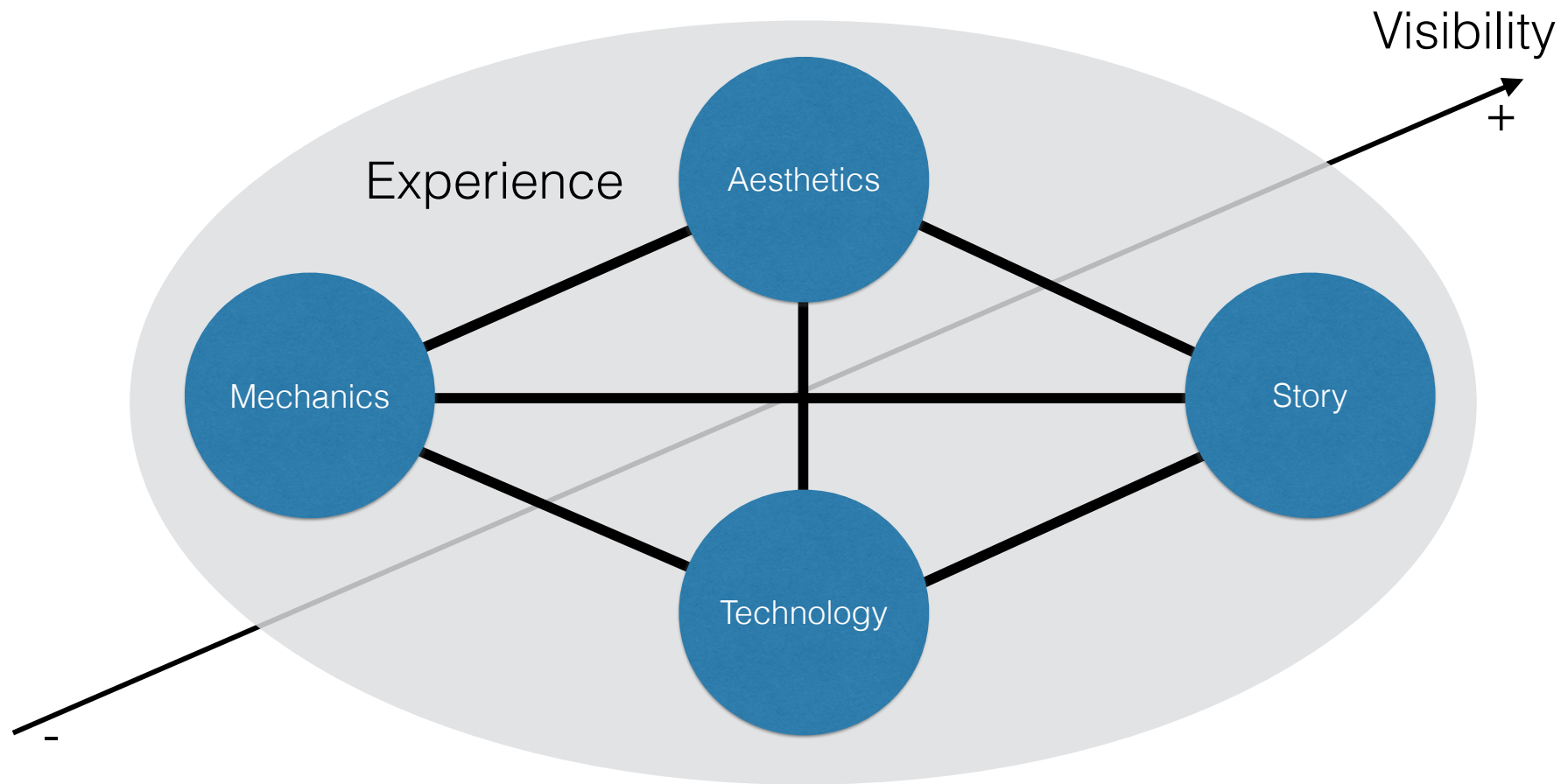
Serious Games

- Are games **just for fun**?
 - There is certainly always a component of entertainment in games but the purpose can be different.
 - Edutainment / Game-based learning
 - Simulation Games
 - Advergames
 - Health Games / Exergames
 - Persuasive Games



[VR Surgeon Simulator]

Game Elements: Elemental Tetrad



Game Elements

- **Mechanics:** These are the procedures and rules of the game. They represent the core interactions and relationships that remain even if the story, technology, and aesthetics are stripped away
- **Story:** This is the sequence of events that unfold throughout the game. It focuses the team's attention and helps humanize the characters to make the game more real and credible for the player
- **Aesthetics:** This refers to the sensory experience—how the game looks, sounds, feels, and even smells or tastes. In the tetrad model, aesthetics are considered one of the "more visible" elements to the player
- **Technology:** This is the medium in which the mechanics occur, the aesthetics take place, and the story is told. It can range from high-end VR hardware to a simple mobile device or console



[The Incredible Machine, 1993]

Ultimately, the goal of balancing these four elements is to create a **State of Flow**—a feeling of energized focus and high fulfillment where the player is completely immersed in the activity. Designers often test their story and mechanics in prototypes using the various elements of this "game tetrad" to ensure the world remains consistent and accessible to the target audience

Game Development

- Game development teams can range from a single developer to large groups where specialisation is king:
 - **Engineers:** develop the software and the tools that make the game (runtime programmers and tool programmers). Further specialisation is possible: rendering, AI, sound, networking, gameplay, scripting, system...
 - **Artists:** produce all the visual and audio content of the game, exception made to procedurally generated content. Specialisations: concept, modellers, texture, lighting, animators, motion capture actors, sound designers, voice actors,...
 - **Game designers:** design the gameplay usually with a hierarchical approach: story arc, high-level goals and objectives, level design and game world areas, challenge design, ...
 - **Producers:** human resources manager, scheduler, interact with marketing and financing areas
 - **Other Staff:** executive directors, marketing, administrative, IT department, ...
 - **Publishers and Studios:** in charge of promoting and selling and distributing the game.

#gamedev

- **Creative Director:** "Yes, we definitely need doors in this game."
- **Project Manager:** "I'll put time on the schedule for people to make doors."
- **Designer:** "I wrote a doc explaining what we need doors to do."
- **Concept Artist:** "I made some gorgeous paintings of doors."
- **Art Director:** "This third painting is exactly the style of doors we need."
- **Environment Artist:** "I took this painting of a door and made it into an object in the game."
- **Animator:** "I made the door open and close."
- **Sound Designer:** "I made the sounds the door creates when it opens and closes."
- **Audio Engineer:** "The sound of the door opening and closing will change based on where the player is and what direction they are facing."
- **Composer:** "I created a theme song for the door."
- **FX Artist:** "I added some cool sparks to the door when it opens."
- **Writer:** "When the door opens, the player will say, 'Hey look! The door opened!'"
- **Lighter:** "There is a bright red light over the door when it's locked, and a green one when it's opened."
- **Legal:** "The environment artist put a Starbucks logo on the door. You need to remove that if you don't want to be sued."
- **Character Artist:** "I don't really care about this door until it can start wearing hats."
- **Gameplay Programmer:** "This door asset now opens and closes based on proximity to the player. It can also be locked and unlocked through script."
- **AI Programmer:** "Enemies and allies now know if a door is there and whether they can go through it."
- **Network Programmer:** "Do all the players need to see the door open at the same time?"
- **Release Engineer:** "You need to get your doors in by 3pm if you want them on the disk."
- **Core Engine Programmer:** "I have optimized the code to allow up to 1024 doors in the game."
- **Tools Programmer:** "I made it even easier for you to place doors."
- **Level Designer:** "I put the door in my level and locked it. After an event, I unlocked it."
- **UI Designer:** "There's now an objective marker on the door, and it has its own icon on the map."
- **Combat Designer:** "Enemies will spawn behind doors, and lay cover fire as their allies enter the room. Unless the player is looking inside the door in which case they will spawn behind a different door."
- **Systems Designer:** "A level 4 player earns 148xp for opening this door at the cost of 3 gold."
- **Monetization Designer:** "We could charge the player \$.99 to open the door now, or wait 24 hours for it to open automatically."

- **Monetization Designer:** "We could charge the player \$.99 to open the door now, or wait 24 hours for it to open automatically."
- **QA Tester:** "I walked to the door. I ran to the door. I jumped at the door. I stood in the doorway until it closed. I saved and reloaded and walked to the door. I died and reloaded then walked to the door. I threw grenades at the door."
- **UX / Usability Researcher:** "I found some people on Craigslist to go through the door so we could see what problems crop up."
- **Localization:** "Door. Puerta. Porta. Porte. Tür. Dør. Deur. Drzwi. Drws. 문"
- **Producer:** "Do we need to give everyone those doors or can we save them for a pre-order bonus?"
- **Publisher:** "Those doors are really going to help this game stand out during the fall line-up."
- **CEO:** "I want you all to know how much I appreciate the time and effort put into making those doors."
- **PR:** "To all our fans, you're going to go crazy over our next reveal #gamedev #doors #nextgen #retweet"
- **Community Manager:** "I let the fans know that their concerns about doors will be addressed in the upcoming patch."
- **Customer Support:** "A player contacted us, confused about doors. I gave them detailed instructions on how to use them."
- **Player:** "I totally didn't even notice a door there."

One of the reasons I like this example is because it's so mundane. There's an impression that game design is flashy and cool and about crazy ideas and fun all the time. But when I start off with, "Let me tell you about doors..." it cuts straight to the everyday practical considerations.

Reposted from: <http://www.lizengland.com/blog/2014/04/the-door-problem/>

The "door problem"

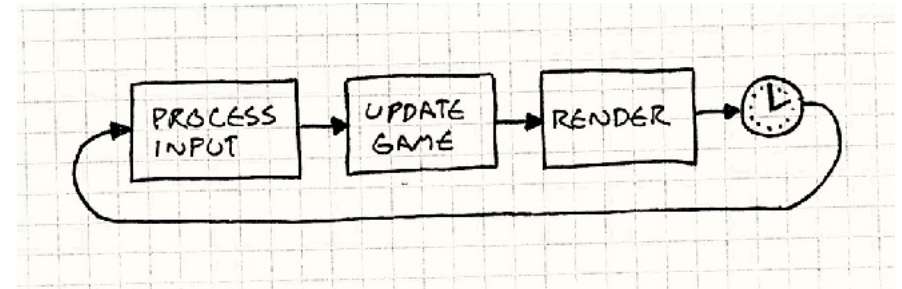
Creating Digital Games

- Exemplo: Cyberpunk 2077, RPG from CD Projekt Red launched in 2020
 - Took 10 years to develop
 - Cost: 274 million euros
 - <https://youtu.be/hZeuG7jI9mE>
- How games are made:
 - UT3: https://youtu.be/R_TU6nHzHDM
 - Open Worlds: <https://youtu.be/K-uBVUGqJss>



The Game Loop

- Interactive applications all need to implement the fundamental loop
 - Take input from user
 - Update the state
 - Render
- Games are *just* another interactive application!



From: <https://gameprogrammingpatterns.com/game-loop.html>

The simulation

- Games are mostly soft, real-time, interactive, agent-based simulations:
 - subset of a real or imaginary **world modelled** mathematically (approximated/simplified) **and simulated** in a computer
 - distinct entities (**agents**) **interact**: vehicles, characters, projectiles, etc.
 - **temporal simulation** with **dynamic content** (user controlled or not)
 - respond to **user input** and **present a story**, thus interactive.
 - soft real-time: a new image every 1/30 ... 1/120 of a second. No disaster if some are generated after the deadline.

Types of Games

We can sort games into categories:

- Action
- Adventure
- Role Playing
- Simulation
- Strategy
- Sports

This previous classification is based on https://en.wikipedia.org/wiki/List_of_video_game_genres

There are other alternative taxonomies based on:

- **purpose** (educational, serious, training, entertainment)
- **type of camera** used (First person, third person, god's view)
- **number of players** (single, two, multiplayer, massive multiplayer)
- ...

Video Game Genres

Action Games

- Based on **physical challenges** requiring **eye-hand coordination**
- Centred around the **player** who is **in control of the action**
- Subtypes:
 - platform games
 - shooter games
 - fighting games

Action Games

Platform games (platformers)

- Platform games (platformers)



Super Mario Bros



Sonic the Hedgehog



Donkey Kong



Manic Miner



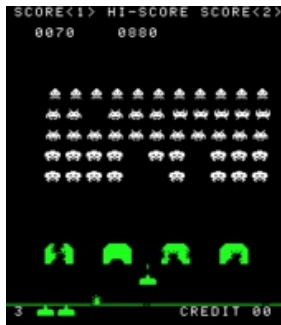
Knight Lore



Little Big Planet 2

Action Games

Shooter games (Shoot 'em up)



Space Invaders



1942

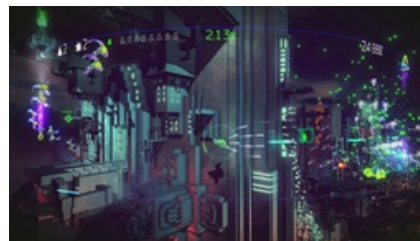


Moon Patrol

- Platform games (aka platformers)



Defender



Resogun



Commando

Action Games

Fighting games



Virtua Fighter



Street Fighter 2



Tekken Tag Tournament

- Platform games (aka platformers)



Fight Night



Mortal Kombat X

Adventure Games

- Some of the first computer games were text based adventure games
- A style of gameplay **without action or reflex challenges**
- Usually some **puzzles need to be solved** by interacting with characters and the environment
- Subtypes:
 - Text adventures
 - Graphic adventures
 - Visual novels and Interactive movies
 - Real-time 3D adventures

Adventure Games



Zork (Text Adventure)



Mystery House



Lords of Midnight

- Platform games (aka platformers)



Shadow of memories



No Man's Sky

Role Playing Games

- The **player** is usually **casted into the role of a specific character**, with **specialised skills**
- The game progresses through a **predetermined timeline**
- Early games were **turn based** games but modern are mostly **real-time** combat games
- Subtypes:
 - Action RPGs - incorporate elements of action or action-adventure games
 - MMORPGs - Massively multiplayer online RPGs
 - Rogue likes - randomised dungeon crawl style games
 - Tactical RPGs - the player controls a finite party and battles a similar number of enemies
 - Sandbox RPGs - large amount of non player characters (NPC) not critical to the game's storyline, large worlds to explore.

Role Playing Games



Diablo (Action RPG)

- Platform (Transformers)



Fable (Sandbox RPG)



Eve online (MMORPG)

Simulation Games

- At the heart of the game is **simulation of a complex phenomenon**
- Subtypes:
 - Construction and Management
 - Life simulation
 - Vehicle simulation

Simulation Games

- Pla



Sim City (Construction)



X-Plane (Vehicle)



Spore (Life)

Strategy Games

- The gameplay relies on **careful** and **skilful thinking** and **planning**.
- Sometimes the player is given a god's view of the game world and controls the units under his command.
- 4 archetypes categorised according to:
 - **turn based** vs. **real time**
 - **military tactics** vs. **strategy**
- TBT, TBS, RTT, RTS

Strategy Games



The Battle of Wesnoth (TBS)

- Platform games (aka platform)



Tacked 3 (RTT)



Starcraft (RTS)

Sports Games

- A special kind of a simulation game where the **simulated object is a sport.**
- Usually **played against** other people or AI
- Common subtypes:
 - Racing
 - Individual/team Sports game

Sports Games

Racing games

- P



Wipeout 2048



Grand Turismo 6

Sports Games



FIFA 2016



PGA Tour

rs)

Game Genres

- The previous classification was based on https://en.wikipedia.org/wiki/List_of_video_game_genres
- There are other alternative taxonomies based on:
 - purpose (educational, serious, training, entertainment)
 - type of camera used (First person, third person, god's view)
 - number of players (single, two, multiplayer, massive multiplayer)
 - ...

Requirements and Features by Genre

Common Requirements

- Process low level user input from devices like keyboard, joystick, mouse
- 3D mesh rendering
- Head-up display - HUD (overlays of text and graphics)
- Audio system
- Resources management
- ...

First Person Shooter Features

- **F**irst **P**erson **S**hooters (FPS) are partially named after the fact that the camera is connected to the head of the player's avatar (as in virtual reality)
- Older games included slow on-foot roaming of a confined rooms and corridors based world.
- Newer games also support vast outdoor areas that are roamed in every imaginable way: by foot, on rails, by vehicle, by ground, water or air

Notable FPS titles

Quake



Notable FPS titles

Unreal Tournament



Notable FPS titles

Half-life



Notable FPS titles

Battlefield 4



FPS Requirements

- Technologically demanding
- Aim to provide the illusion of being immersed in a detailed and realistic world
- Many technological advances came from this type of games
- Requirements:
 - efficient rendering of 3D virtual worlds
 - responsive camera control and aiming
 - high-fidelity animations of the player's arms and weapons
 - powerful handheld weaponry
 - forgiving player character motion and collision model (floaty feel)
 - high-fidelity animations and artificial intelligence of non-player characters
 - small-scale online multiplayer capabilities

Platformers and Third-Person*

Features

- Platformer is a third-person character-based action game where jumping from platform to platform is the primary gameplay mechanic
- The camera follows the character
- The game action can happen in a 2D, 2.5D or a 3D world.
- Platformers and third-person shooters/action/adventure share requirements.
- A lot of common requirements from First Person games
- More emphasis on main character's abilities and locomotion (full body animation)

Notable platformer/TP* titles

New Super Mario Bros



Notable platformer/TP* titles

Super Mario 64



Notable platformer/TP* titles

Crash Bandicoot



Notable platformer/TP* titles

Jak and Daxter



Notable platformer/TP* titles

Little Big Planet



Notable platformer/TP* titles

The Last of Us



Notable platformer/TP* titles

The Last of Us



Platformer and Third-Person* Requirements

- moving platforms, ladders, ropes, etc.
- puzzle-like environmental elements.
- third-person “follow camera” focused on the player (in 3D it can rotate)
- camera collision system to ensure that the camera is not affected by foreground objects.

Fighting Games

Features

- Typically two player games with humanoid characters
- Small game world (usually a ring, patio, etc.)
- Detailed movements
- Complex blows and attacks
- Different characters with different abilities/skills

Notable Fighting Games

Virtua Fighter 5



Notable Fighting Games

Fight Night 4



Fighting Games Requirements

- Rich set of fighting animations
- accurate hit detection (an ellipsoid around the body won't work)
- user input system capable of detecting complex button combinations (and more important: sequences of input)
- crowds with relatively static backgrounds
- high-definition character graphics, realistic skin shaders, subsurface scattering, sweating, bruises, ...
- physics based cloth and hair animation

Racing Games

Features

- Goal: Drive a car or other vehicle in some kind of track.
- Some, like Grand Turismo, focus on realism and are labeled as sims, while others (e.g. Sega Rally Championship) explore the fun and entertainment (arcade style).

Notable Racing Games

Sega Rally



Notable Racing Games

Grand Turismo



Racing Games Requirements

- tricks for rendering distant objects like trees and mountains
- track needs to be broken down into sectors to optimize rendering (visibility determination)
- camera can be outside (third person) or in the cockpit (fps) or even in the public or in an helicopter (replays)

Real-time Strategy Games Features

- Player deploys the battle units across a large playing field
- Playing field displayed in an oblique top-down viewing angle
- No long distance viewing which leads to some optimisations
- Older games used cells in a grid layout to align elements like buildings properly
- Player's orders are translated into micro actions

Notable RTS Games

Dune II: The Building of a Dynasty



Notable RTS Games

Command & Conquer 3



Notable RTS Games

Command & Conquer 3



Notable RTS Games

Command & Conquer 3



Real-time Strategy Games Requirements

- Each unit has a relatively low resolution to support a large number of them simultaneously
- The game world is usually an height-field terrain
- Apart from deploying the units, building constructions is also usually allowed
- user interaction is single click (single selection) or area based (multiple selection) followed by menus, toolbars, shortcuts.
- The game simulates order execution by having agents performing micro level tasks: move, aim, fire, retreat, avoid, etc.

MMOG* Games

Features

- Any game supporting from a thousand to hundreds of thousands of simultaneous players
- Usually all players are playing in a large persistent world (persistent across individual game sessions)
- A battery of servers at the back of the game (sign-in, sign-out, subscriptions, keep authoritative state of the world, handle micro-transactions)
- Subtypes: MMORPG, MMORTS, MMOFPS
- Similar to non MMOG* counterparts but with lower graphics fidelity to be able to handle huge world sizes and large number of players

Notable MMOG* Games

World of Warcraft



Player-Authored Content Games Features

- Recent trend in game industry as social media take off
- Encourage players in creating, polishing and sharing game content (levels, characters, etc.)
- Reduces the need for extension packs and allow longer longevity

Notable Player-Authored Content Games

Little Big Planet/LBP2/LBP3

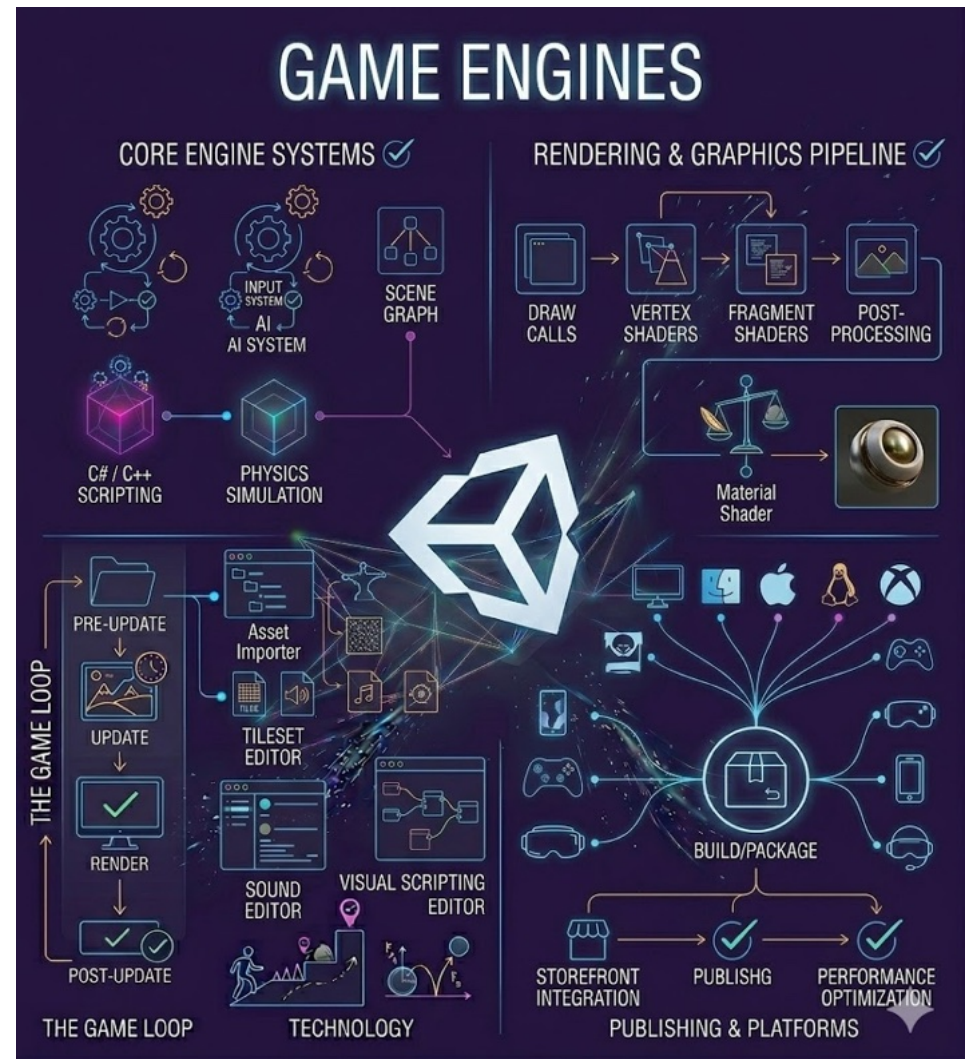


Notable Player-Authored Content Games

Minecraft



Game Engines



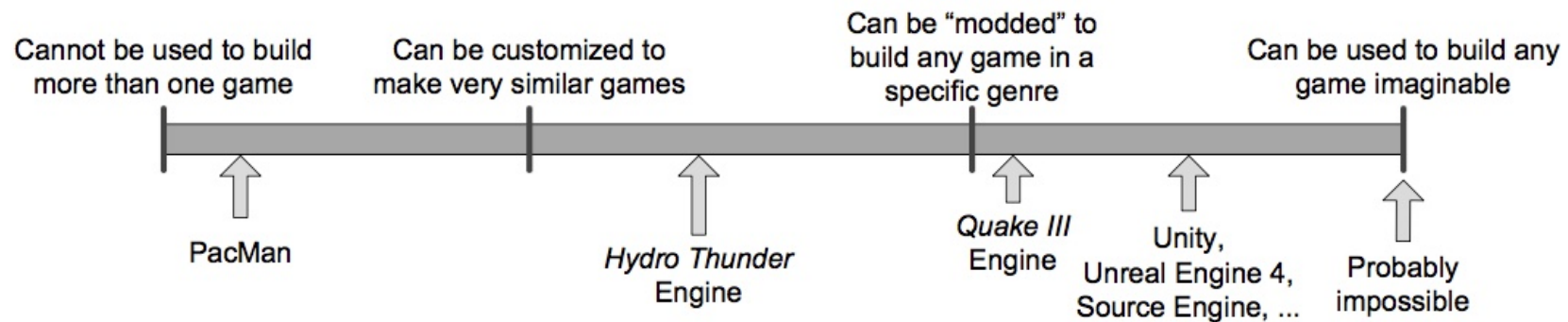
Why use an Engine?

- Nowadays, no one programs a game from scratch!
 - In the early years, a game was programmed in assembly language and tied to a specific hardware
 - slow development, few highly skilled developers
 - Over time, games started to fall into categories, bearing resemblances in style and gameplay with other titles
 - Eventually, development teams started to put the common features into game libraries that could be reused in different/similar games: rendering, networking, input, AI techniques, ...
 - Shorter development period, more programmers could build a game
 - Libraries later gave way to entire development suits composed of realtime game engines with scripting to avoid low level programming and other development tools such as game world editors, animation editors, sound editors, etc.
 - everyone can create a game!

What is a game engine?

- It is composed of several subsystems that deal with specific parts/needs of a game:
 - graphics/rendering
 - scene/world management
 - animation
 - sound
 - networking
 - user interaction
 - operating system interaction
 - ...
- A game engine is just a part of the overall suite of tools used in game development.
- It is the part that interests engineers the most...

Game engine reusability



Game engine reusability gamut

Reproduced from Jason Gregory's "Game Engine Architecture", 2nd Ed.

There is no universal game engine yet...

Game Engines 2D/3D

- Some game engines are oriented to 2D sprite based games
- A sprite is a rectangular array of pixels that can be manipulated on the screen
- Old video games were sprite based
- Old hardware had sprite specific hardware (bitmap manipulations)
- Today, graphics hardware is 3D oriented. 2D worlds a special case of 3D worlds (Flat objects, an orthographic camera, and textures...)



Core Architecture of Game Engines

- **Scene Graph:** How data is managed and organized in 3D space
- **Rendering Engine:** Managing the 3D graphics pipeline and hardware
- **Physics Engine:** Handling collisions and numerical methods
- **Scripting & Logic:** How game mechanics are implemented

Immersive Environments



Virtual Reality (VR)

Virtual Reality is a real-time, interactive, computer-generated simulation of a three-dimensional environment that supports user immersion and sensorimotor interaction, producing a sense of presence within the simulated world.

Core Characteristics:

- **Immersion**

The user is perceptually surrounded by the virtual environment, typically through a head-mounted display (HMD) that blocks out the physical world.

- **Interactivity**

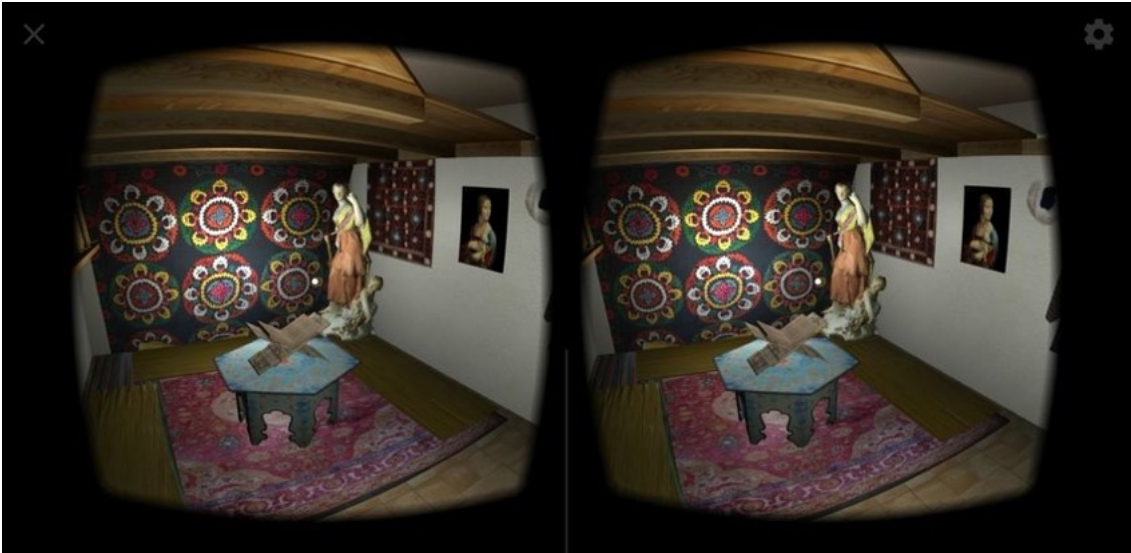
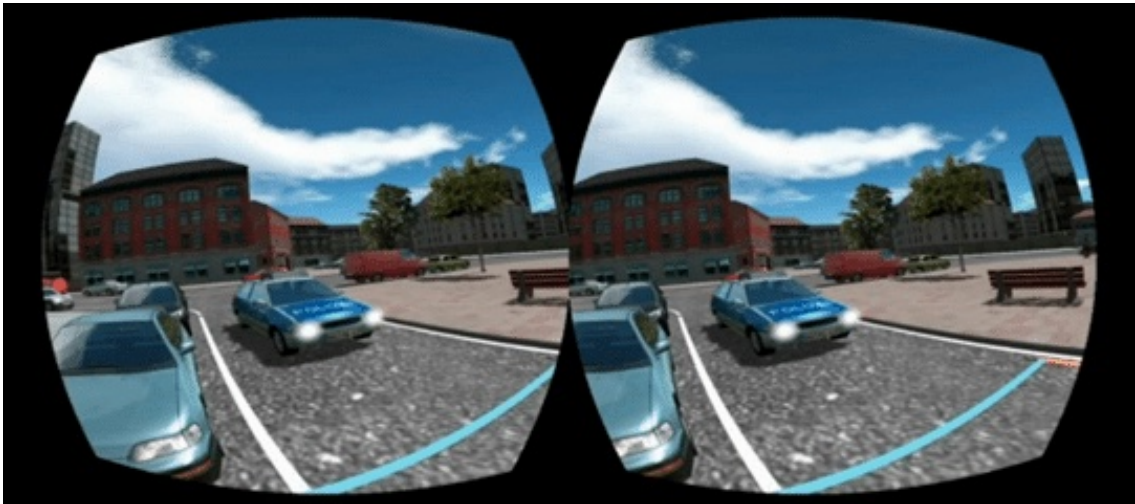
The system responds in real time to the user's movements and actions (head, hands, body).

- **Real-time Rendering**

The environment updates continuously based on user input.

- **Sense of Presence**

Users experience the psychological feeling of “being there” inside the virtual space.



Augmented Reality (AR)

- Augmented Reality (AR) is based on the principle of augmenting — not replacing — the physical environment by superimposing computer-generated content in real time, spatially aligned with the real world.:
- A widely accepted theoretical framework comes from Ronald Azuma (1997), who defined AR systems as those that:
 - Combine real and virtual content
 - Are interactive in real time
 - Are registered (aligned) in 3D
- The key concept here is **registration** — digital content must appear anchored to the correct physical location.

Spatial Registration

- For digital objects to appear “on” the real world:
 - The system tracks camera/head position (pose estimation)
 - It builds a model of the physical environment
 - Virtual objects are rendered using the same perspective geometry
- This relies on:
 - Computer vision
 - SLAM (Simultaneous Localization and Mapping)
 - Sensor fusion (camera + IMU + depth sensors)
- Without correct registration, the illusion breaks.



Team planning

This month

	Owner	Timeline	Status	Date
Finalize kickoff materials			Done	Sep 04
Refine objectives			Working on it	Sep 09
Identify key resources			Done	Sep 15

Next month

	Owner	Timeline	Status	Date
New candidate interviews			Working on it	Oct 06
Lead generation presentation			Done	Oct 10
Optimize campaign spend			Stuck	Oct 12
Meeting with publishers			Working on it	Oct 17

Weekly Team Sync

Agenda

1. Kickoff & team reflection
2. Brainstorm
3. Action items



Brainstorm



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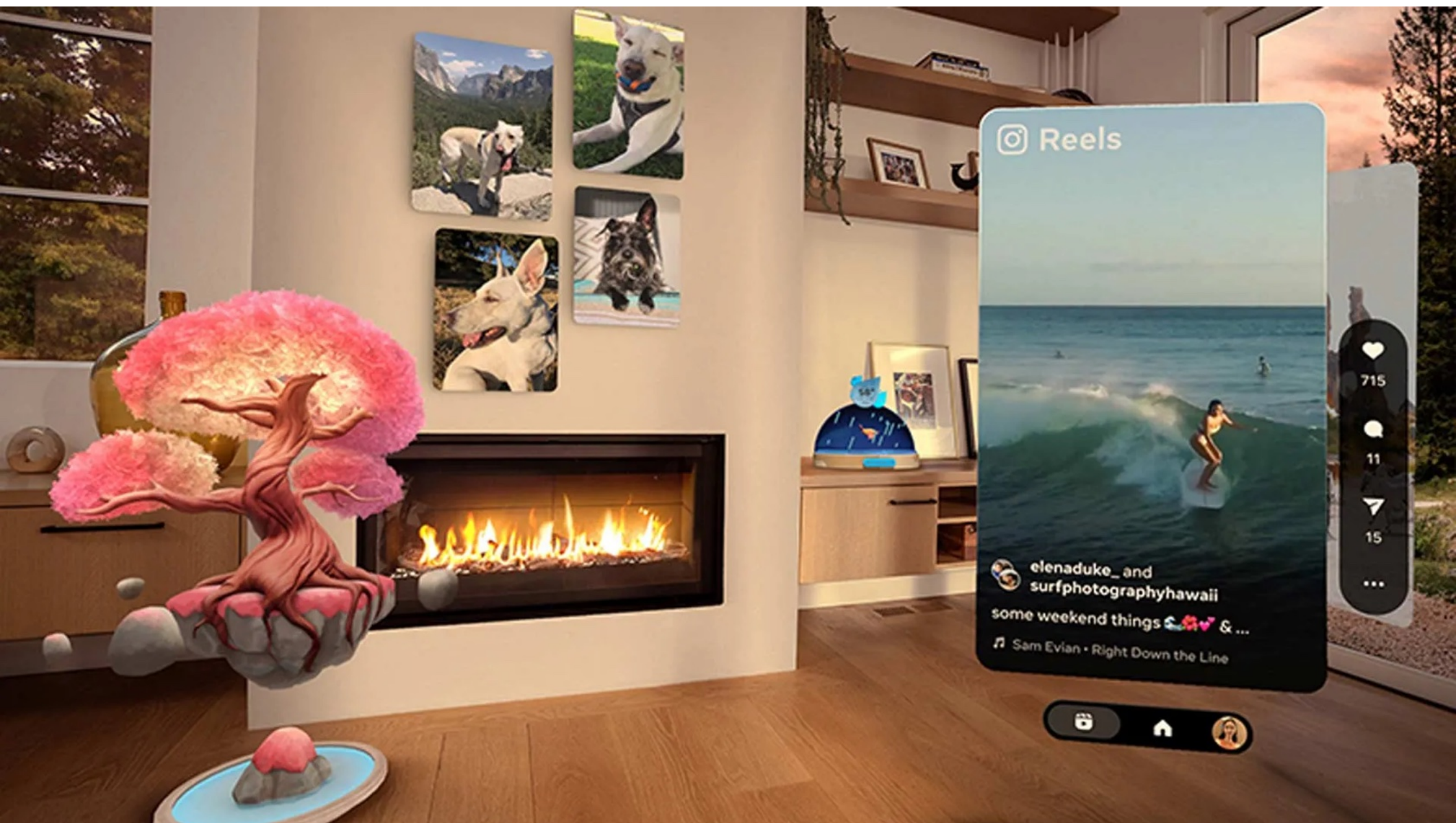
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The Hardware

- Brief mention of current graphics hardware and systems

Demos